Somatic variant calling & Annotation

Massimo Maiolo

Github Cancer Genomics Course



https://github.com/supermax1234/CancerGenomicsCourse

Lesson plan

17.09.2026	Panos - Sofia	Introduction/Logistics + Introduction to cancer biology
24.09.2026	Philip	Introduction to software for mutation calling, gene signatures
01.10.2026	Sofia	Cancer Biology
08.10.2026	**Max**	**Somatic variant calling**
15.10.2026	**Max**	**Somatic variant calling + variant annoation**
22.10.2026	Philip	Variant prioritization - Driver mutations
29.10.2026	Philip	Mutational Signatures
05.11.2026	Andrew	Mid-term exam
12.11.2026	Panos - Andrew	Introduction to software for structural analysis
19.11.2026	Panos - Andrew	CNV calling
26.11.2026	Panos - Andrew	Structural variants
03.12.2026	Panos - Andrew	Tumor clonality
10.12.2026	Panos - Andrew	Data interpretation + summary
17.12.2026	All	Summary - Q&As
13.01.2027	Panos - Andrew	Final Exam

Course Assignment: Group Organization

Teamwork & Structure

- Organize yourselves into **groups of 3 students** (10 groups total)
- **Work Together:**
  - Team members share the exact same grade for this assignment, so collaborate closely,
  - review each other's scripts, and
  - run through the pipeline as a team
- Please provide the names of the group members asap

Course Assignment: Submission Guidelines

Submission Details

- **Format**: Submit (email: massimo.maiolo@unibe.ch) a single compressed *.zip* folder containing:
  - all your developed SLURM scripts, following the sequential numbering convention (e.g., *01_*, *02_*, etc.) covered in class
  - A **README.md** text file containing the full names and surnames of the 3 group members. Any notes, comments, or explanations in the README if a specific step failed, threw warnings, or required alternative handling
- **Deadline**: submit your group's work by **October 25**

Important Reminder: SLURM Script Requirements

SLURM scripts vs. Local bash

- Every script you submit must be a fully functional, runnable **SLURM batch script**, not a local bash script
- Each script must start with the required *#SBATCH* header (partition, memory, time, job name, and reservation) as shown in the course template
- **Never** run heavy compute tasks directly on the master/login node; always submit your scripts via sbatch